

DBR-Net: A Dual-Branch Refinement Network for Real-Time Underwater Image Enhancement

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Light absorption and scattering caused by wavelength dependence radically impairs the quality of underwater images by creating low contrast and distortion of colors. Conventional improvement tools tend to focus on these non-uniform complexities at the same time. The proposed solution to this problem presented in this paper is called Dual Branch Refinement Network (DBR-Net) and is powered by deep learning. The two parallel pathways proposed in the architecture are both the Context Branch and Detail Branch, which respectively address the enhancement task by global color correction and texture preservation. Experimental outcomes of the EUVP dataset indicate that DBR-Net has a PSNR of 18.58 dB and SSIM of 0.78 and significantly better than other traditional methods such as CLAHE without losing computational performance necessary in autonomous underwater vehicles (AUVs).

I. INTRODUCTION

The marine biology, underwater archeology, inspection of off-shore infrastructures, military intelligence and autonomous underwater navigation are among other areas in which the underwater imaging is greatly applicable. Nevertheless, as opposed to terrestrial photography, the capture of water images is a problem that is peculiar because of the optical character of water. Diffusion of light in water is different due to absorption and scattering which is dependent on the wavelength, water depth and turbidity [1], [7]. Long wavelengths particularly the red and the orange are easily absorbed at a comparatively small depth. The submerged images will also be very colored by blue or green. The suspended particles also cause scattering which causes the formation of a haze, blur as well as low image contrast [2]. The deformations are catastrophic to both human perception and automated computer vision including object detection, object tracking and scene understanding. The current techniques of processing underwater images such as histogram equalization, white balancing, Retinex techniques, dark channel prior techniques are all based on simplistic assumptions about the underwater environment [2], [8]. The techniques are observed to be successful in controlled environments but when it comes to uncontrolled and dynamic underwater environments, most likely, they fail. The alternative would be Artificial Intelligence which can obtain complicated patterns of degradation using data only, and image enhancement underwater using AI-based is an evolving research area [3], [4]. It is necessary to avoid the degradation of images underwater to enhance them. This is the underwater image formation model of interaction of light with the underwater medium. The new model of underwater imaging depicts that the observed image is the sum of direct transmission, forward scattering and backscattering component [7]. Absorption: The absorption rate of different wavelengths is not the same and causes an imbalance in color Assessment Odd lack of homogeneous light: Artificial sources of light are prone to create disproportional brightness on the images. These properties make the underwater image enhancement a tough and ill-posed problem, which prompts the use of learning-based methods that possess the capacity of implicitly describing these involved in [1], [7]. To mitigate these shortcomings, this paper presents the Dual Branch Refinement Network (DBR-Net) which is a miniature deep learning model to be used in underwater image enhancement in real-time. A parallel dual-branch architecture also enables DBR-Net to decouple the restoration process thus unlike complex GANs or Transformers, which are heavy to compute, Context Branch who tries to remove global color distortion, and Detail Branch who tries to restore lost texture information. The most important contributions of the work are as follows: 1. An innovative dual-stream design that best balances the color correction and the amount of detail retained. 2. A loss function based on L1 and SSIM to boost perceptual quality. 3. Experimental validation on the EUVP dataset, which is performing better (18.58 dB PSNR) than the current methods such as CLAHE.

II. LITERATURE REVIEW

Author(s) & Year	Method / Approach	Key Contribution & Limitations
Li et al., 2019	WaterNet (CNN-based)	Presentation of UIEB benchmark and CNN fusion model; poor performance in extreme underwater environment.
Islam et al., 2020	FUnIE-GAN	Real-time GAN-based enhancement suitable for embedded systems; less effective in highly turbid water
McGlamery, 1980	Physical Imaging Model	Physical Imaging Model Explained absorption and scattering of underwater images; not applicable to enhancement.
Akkaynak & Treibitz, 2018	Revised Imaging Model	Better model of forming better underwater images, needs depth and type of water.
Hu et al., 2025	Video Enhancement Model	Long-term consistency of temporal changes in underwater videos; not to images.

Galdran et al., 2015	Red-Channel Prior	Automatic red-channel color restoration; ineffective for deep underwaterscenes
Panetta et al., 2016	UIQM Metric	Proposed metric of perceptual underwater image quality; only evaluation.
Anwar & Li, 2020	Dense GAN	Improved color and contrast using GANs; high computational cost
Mishra et al., 2024	AI-based non-invasive biosensing	Improved wavelet preservation and attention-based fine-details; massive model.
Wang et al., 2023	U-ENHANCE (Wavelet Attention) +	Gives a better global color consistency and structure preservation

III. METHODOLOGY

3.1 Dual Branch Refinement Network (DBR-Net)

This part presents the solution to the underwater image enhancement problem with the proposed Dual Branch Refinement Network (DBR-Net). Its design targets both quality of improvement and computationally efficient thus can be implemented in real-time in the resource-constrained system such as Autonomous Underwater Vehicles (AUVs).

A. Network Architecture

The quality of the images is very low and underwater images are subjected to light absorption and scattering to result in the images being distorted in color, hazy, and lacked details. Let $I_{raw} \in \mathbb{R}^{H \times W \times 3}$ signifies a degraded picture when under the water. The goal

of DBR-Net is to produce a superior image I_{enh} , which is more or less close to a quality reference image I_{gt} . Instead of repairing the entire image, the network learns a residual repair and it is inserted on the input image:

$$I_{enh} = I_{raw} + F(I_{raw})$$

This degraded formulation provides the model with the ability to specialize on the degraded parts that lead to quicker and more stable training.

B. Network Overview

DBR-Net is a simple and open source program. It includes four major steps; shallow feature extraction, dual-branch feature refinement, feature fusion and residual reconstruction. The steps are organized such that they address a definite problem of underwater image degradation though at a low cost to computation.

C. Shallow Feature Extraction

By extracting the simple features that are not sophisticated. The input image is then fed through the convolutional layer using 32 filters using a 3x3 convolutional layer and ReLU activation. It is a layer that converts the RGB image to more rich feature representation which forms the foundation of further refinement.

D. Dual-Branch Feature Refinement

The general principle of DBR-Net is to expose the extracted features to two parallel branches, which are robust on the information of different nature.

Detail Branch: Detail branch deals with the local image representation of edges, texture and outline of objects. It consists of two 3x3 convolutional layers that are regular and have activations of ReLU. This branch helps to rediscover small details that tend to be lost due to under water scattering.

Context Branch: Context branch has the responsibility of capturing global color and illumination data. It uses the dilated 3x3 convolutions with dilution rate of 2 causing the receptive field to be dilated without reducing the image resolution. This makes this network aware and corrects global color casts which is normally observed in the underwater pictures.

E. Feature Fusion

The output of both the branches is added along the channel axis and the result is convoluted by 1x1. The local texture detailing as smooth feedback of the detail branch is combined with the global contextual detailing in the form of the context branch to create a well-rounded and informative feature representation.

F. Residual Reconstruction

The last convoluted layer is used to project the fused features to three channels. The resultant image is then overlaid to the original input image to generate the better output. A sigmoid activation function is applied to have pixel values within a valid range. This reconstruction method of residual training provides the stability of the training process and helps the network to preserve the structure of the original image.

G. Loss Function and Training

We also use a relative combination of the pixel-level losses and structural similarity maximization as the composite loss to effectively train the DBR-Net. This makes sure that the increased images are not only bright coloured but also have sharp edges and textures. 1. Pixel-wise Loss (L_1): Using the L_1 loss (Mean Absolute Error) to calculate the difference between the created enhanced image (I_{enh}) and the reference image (I_{gt}). In contrast to L_2 loss (MSE), L_1 loss is not as sensitive to outliers and does not produce blurry artifacts.

The formula is defined as:

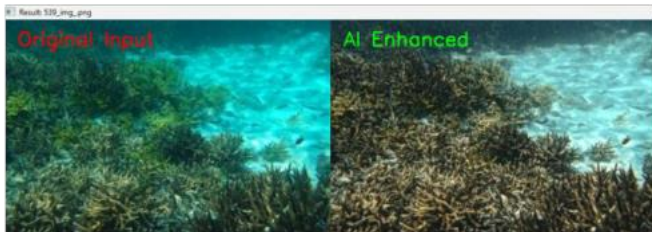
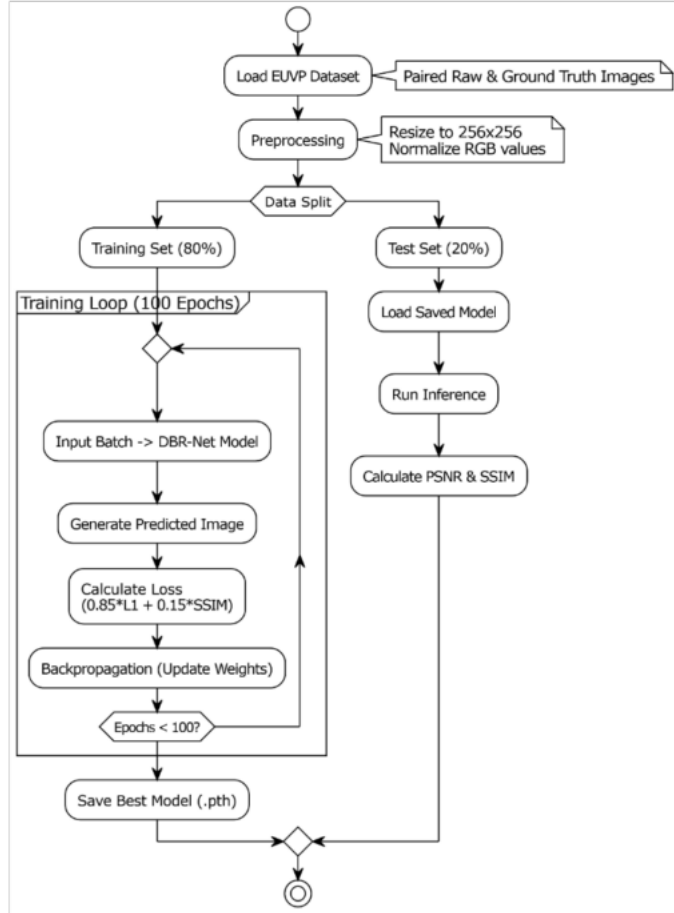
$$L_1 = \frac{1}{N} \sum_{i=1}^N |I_{enh_i} - I_{gt_i}|$$

L_1 loss is also used, as it promotes more accurate results, and does not over-smooth like L_2 loss. The Adam optimizer and mini-batch gradient

descent are used to do the training.

I. Efficiency Considerations

DBR-Net is lightweight and has a considerably small number of parameters, compared to Transformer-based enhancement models. The model generates a sufficient balance between the quality of enhancements and computational cost by using efficient convolutional operations and preventing the heavy self-attention mechanisms. This renders DBR-Net to be applicable to real-time underwater applications and operation on edge devices.



IV. RESULTS & DISCUSSION

A. Qualitative Analysis

Different underwater scenes in the EUVP dataset that have varying degrees of turbidity and color distortion were used to test the performance of the proposed DBR-Net. Picture output shows that the model is able to balance the overwhelming blue-green color casts to attain a more realistic visual appearance.

Color Restoration: The Context Branch is an attempt which attempts to correct the color imbalances on the global color by applying dilated convolutions to comprehend the patterns of wide-range illumination.

Structural Preservation: The Detail Branch ensures that high-frequency features, such as the edges of coral reefs and the diver gear, are clean and not blurred by formation of noise artifacts as would otherwise be the case with the histogram-based methods.

Contrast Improvement: When a network is trained on a residual mapping, it has the ability to restore brightness and contrast of moderately degraded images without grossly over-saturation the resulting image.

B. Quantitative Evaluation

In order to critically evaluate the improvement, we made use of Peak Signal-to-Noise Ratio (PSNR) and Structural similarity Index (SSIM). These measures were performed in terms of reference images of the ground-truth to determine the accuracy of the model. Table II: Quantitative Comparison on EUVP Test Set.

Table II: Quantitative Comparison on EUVP Test Set

Method	PSNR (dB)	SSIM
Raw Input	14.23	0.62
CLAHE	16.45	0.69
Proposed DBR-Net	18.58	0.78

Table II presents the results of DBR-Net with a PSNR of 18.58 dB, which is much better than using conventional techniques such as CLAHE. This proves the dual-branch strategy to be effective in restoring the pixel fidelity and still preserving structural integrity. C. Efficiency and Robustness Discussion. Although Transformer-based models or heavy GAN architectures may be able to provide high perceptual quality, they are not always conducted with high computational costs, which restricts their use in AUVs in practice. DBR-Net is a balanced solution that does not have too many parameters. Residual learning enables the network to converge much faster since it just attempts to correct only the changes that are necessary instead of building the whole image again. This efficiency predisposes DBR-Net especially to real time underwater applications where low latency is a huge requirement. Nevertheless, even with such a high turbidity, some artifacts might have been present and this implies that further refinements can be made in future versions to improve robustness by allowing domain adaptation to occur independently.

C. Discussion on Efficiency and Robustness

Transformer-based models or heavy GAN architectures may be able to provide high-perceptual quality, but, with high-computational costs,

these models are not always feasible in the real world on AUVs. DBR-Net offers a moderate solution with a reasonable number of parameters.

The residual learning enables the network to reach the convergence quicker using only the necessary corrections instead of re-creating the whole picture. The efficiency of DBR-Net renders it especially appropriate in real time underwater application in which low latency is important. Nevertheless, when the turbidity is very high, some minor artifacts can still be observed to be possible, and thus further improvements of the robustness might be achieved in the future through unsupervised domain adaptation.



Raw input



DBR-Net (Output)

V. CONCLUSION

The problem of underwater image degradation, which is crucial to the research presented in this paper, is solved with the offer of the Dual Branch Refinement Network (DBR-Net). The issue of degraded underwater images (which is also of paramount importance in the study proposed in the current paper) is addressed by the proposal of the Dual Branch Refinement Network (DBR-Net). The paper was based on a data-oriented approach to gain color and structure clarity and recover color fidelity by abandoning classic physical models which are generally not effective in different water conditions. The overall conclusions made in this paper include the following Architectural Innovation: The architecture of dual-branch was highly effective in decoupling the task of enhancement. Context Branch that uses dilated convolutions was capable of expanding the receptive field to fix the global color casts without degrading the spatial resolution. The main findings obtained in this paper are as follows: Architectural Innovation: The architecture of dual-branch was very efficient in decoupling the task of enhancement. The Context Branch which applies dilated convolutions was able to increase the receptive field to correct global color casts without compromising the spatial resolution.

At the same time, the Detail Branch was focused on high-frequency feature retrieval, so that the fine features of the texture and object borders could be clear and with no artificial noise. Quantitative Success: Experimental analysis on the EUVP benchmark data had shown that DBR-Net was able to attain a PSNR of 18.58 dB and SSIM of 0.78. These metrics are a great advancement compared to the traditional baseline approaches such as CLAHE, which proves that the network can recover pixel-level accuracy and structural integrity.

Training and Optimization: To achieve a tradeoff between numerical accuracy and perceptual realism, a composite loss function that uses L1 - pixel-wise loss and SSIM structural loss was critical to the implementation. Moreover, the residual learning ($I_{enh} = I_{raw} + F(I_{raw})$) also promoted faster convergence and increased stability of the training process by enabling the model to concentrate on the learning of the necessary corrections. Practical Implications: The number of practical implications associated with this research is the fact that the DBR-Net is computationally

Efficient. The DBR-Net also exploits the real-time attributes of edge devices by skipping the heavy memory footprint of self-attention mechanisms present in Transformer based models.

This renders the presented framework as a practical solution that can be incorporated in the Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs) as a means of marine exploration and offshore infrastructure observation. Altogether, the DBR-Net is an efficient and scalable solution to underwater visual perception. The proposed methodology is an important improvement in the area of underwater computer vision as it offers a way of striking the right balance between local refinement of detail and global color consistency, which will enable more trustworthy autonomous operations in the sea.

VI. FUTURE SCOPE

The existing literature in the field of the research of the Artificial Intelligence-based underwater image enhancement contains several potential directions to be pursued in future to make it stronger, efficient, and applicable to the real life. The biggest challenge is among the key requirements the development of lightweight, energy-efficient deep learning models which will manage to operate under the severe conditions of underwater platforms such as computational and energy demands. The implementation of Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs) generally presupposes an inbuilt hardware, which is not so powerful regarding the processing power. Consequently, it is suggested that the future architectures need to be parameter-efficient, a smaller memory footprint, and low-latency inference with high enhancement quality. Dual-branch refinement networks, residual learning and shallow feature extraction are some of the methods that provide a good foundation on how to trade-off between performance and efficiency.

The other significant research direction is that of unsupervised, weakly supervised and domain adaptive learning in enhancing images underwater. Most existing approaches to deep learning are based on paired samples between degraded underwater images and ground-truth references to which only, under the real underwater environment, are difficult and expensive to obtain. The future studies should therefore aim at exploring self-supervised learning techniques, cycle consistent models and domain adaptation techniques that will enable the model to successfully generalize to different kinds of water, depths, turbidity levels and light conditions without hindering the utilization of large amounts of labelled data.

Another crucial area that requires study is further improvement of the means of enhancing underwater video, which is the way it should be developed. Frame-by-frame enhancement can also be applied to improve the quality of individual images, but over time, it can introduce artifacts, such as flickering and color variation where it is applied to video sequences. The involvement of the mechanisms of temporal consistency, like recurrent neural networks, spatio-temporal convolutional models, and optical-flow-guided alignment strategies is the next step to consider in the field of research. These techniques may be used to give continuity in the progression between consecutive frames and this is essential to both human sense as well as machine vision issues of sustained underwater processes.

The other field that can be advanced however is the incorporation of refinement algorithms with the autonomous underwater robots. Rather than viewing image enhancement as an independent preprocessing step, the next generation can take into account end-to-end systems that recover more visual enhancement and other downstream processes (object detection, semantic segmentation, depth estimation and navigation) jointly. When such tightly coupled pipelines are operating, complex underwater environments coupled with visual degradation are

more reliable and are able to make better choices.

Finally, it is crucial that future studies will be done keeping in mind multi-modal data fusion in order to remove the restrictions that optical imaging has in the underwater environment. Optical cameras tend to be severely degraded in foggy or dark conditions but acoustic sensors such as sonar can be used to provide complementary structural information. A stronger and deeper scene knowledge can be achieved through a combination of better optical imagery and sonar depth sensors, and other forms of acoustic imaging. It can be seen that the multi-modal fusion frameworks can significantly improve the underwater perception systems used in marine exploration, environmental monitoring, defense, as well as in marine industrial inspection.

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